

# ccint — Progress Brief

Canada Cyber Social Intelligence: design rationale, build, and where it stands

22 September 2026

## The question, and what changed about it

The project began as a longitudinal study: *how does Canada-related cybersecurity discussion change over time on social media?* A collector runs continuously against Bluesky, posts are labelled under versioned schemes, and change is tested statistically.

The first full analysis returned a negative result, and returned it with enough rigour to be worth something. The largest topic movement in the 30-day window, `data_breach` at +22%, has  $p_{\text{FWER}} = 0.734$  under an author-level circular-shift null that preserves each author’s volume, topic mix and burstiness while destroying calendar alignment. Empirical power analysis — measured by author-pool replication rather than assumed — puts the requirement at roughly  $8\times$  the current volume.

That result is only interesting if one knows what produced it. Three explanations were open, and nothing in the system could separate them, because every figure it produced came from labels it had produced itself. The work of the last phase was to build the measurements that can separate them. They do, and the answer was not the expected one: **the shortfall is not in the statistics and not mainly in the labelling — it is that the platform carries almost no human discussion of these events at all.**

## Why the system is built this way

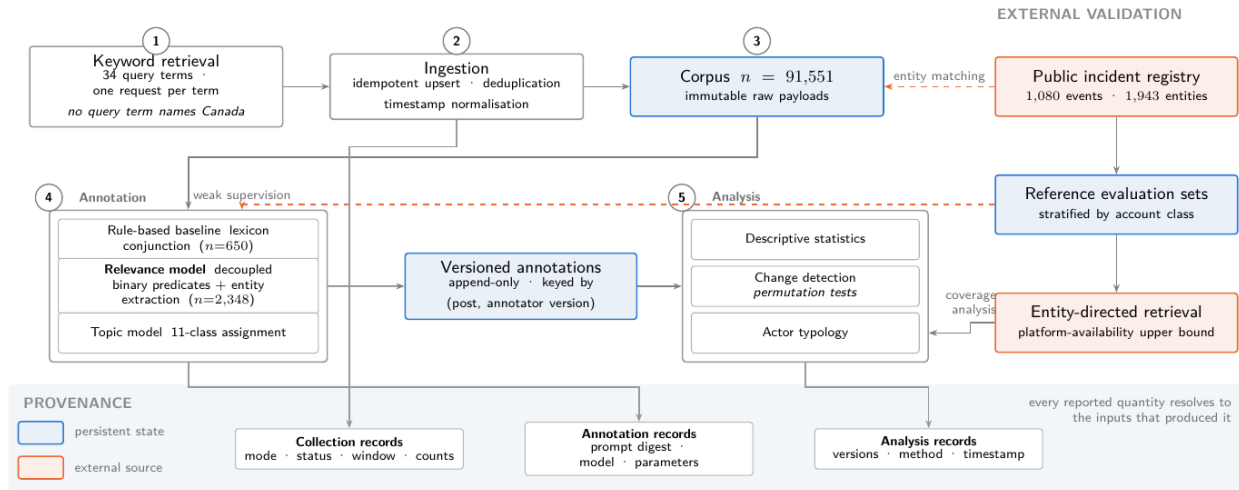
Four commitments shaped every design decision. Each exists to prevent a specific failure that would be invisible after the fact.

**Bluesky, and one platform only.** An open, documented API with usable historical search and terms that permit research collection; X and Reddit were excluded on access grounds rather than merit. The consequence is stated rather than hidden: this is a convenience sample, and the project’s job became partly to measure what it can show.

**Labels are versioned and never overwritten.** Annotation rows are keyed by (post, annotator version) and only ever inserted; four schemes coexist over the same corpus and diff row by row. Without this, every improvement to the annotator would silently rewrite history, and a shift in topic distribution could never be attributed to the world rather than to the method.

**Raw data is kept verbatim; every number carries provenance.** `raw_payload` is stored unmodified, and every run and model label records what produced it — query version, label version, method, and for model labels the SHA-256 of the exact prompt text. The last was added recently, after noticing that a model labeller’s output is entirely determined by its prompt: “labels are never overwritten” had been preserving the labels while losing the conditions that made them.

**System failure must be distinguishable from social change.** Every collection path records mode, status, window and counts. A day with half the usual volume must be attributable to either the world or a token expiry, and only bookkeeping makes that possible. The principle held under test: a 42-run token outage appears in the record as 42 `partial` runs, not as a drop in Canadian cyber discussion.



**Figure 1:** Processing pipeline. Stages 1–3 acquire and store the corpus, stage 4 annotates it under coexisting versioned schemes, stage 5 analyses the result. The external-validation column supplies labels no component of the pipeline produced.

Underlying all four: **measure before improving**. Repeatedly the intuitive next step turned out to be aimed at the wrong quantity, and only measurement revealed it — tuning the trend score would have been wasted, relaxing the relevance rules helps but not for the population that seemed to need it, and the most recent prompt improvement introduced a regression the evaluation set certified as correct.

## How it was built

Collection runs in two modes against the fixed term list — a day-by-day backfill and an hourly incremental — with idempotent ingestion, stale-run reaping and client-timestamp repair. Annotation proceeded in four generations, all retained: two lexicon-based, one that re-decides topic only while inheriting the relevance verdict verbatim so the two changes stay separately attributable, and one that replaces the relevance decision entirely. Analysis covers descriptive statistics, permutation testing against two null models, behavioural broadcast detection with a collection-lag control, and a two-axis actor typology separating measured audience from content-judged function. The most recent phase added the external-validation column of Figure 1.

## Where it stands

### The collection boundary was never measured

The 34 search terms came from a specification naming four examples followed by “etc.”; the remaining 26 were supplied during implementation from domain intuition, never derived from data and never revised. Reconstructing per-term productivity offline over the stored corpus:

| Search term      | Posts retrieved | Judged relevant |
|------------------|-----------------|-----------------|
| cybersecurity    | 22,498          | 200             |
| ransomware       | 8,881           | 105             |
| data breach      | 1,939           | 90              |
| CVE              | <b>12,502</b>   | <b>3</b>        |
| 11 further terms | —               | <b>0</b>        |

CVE accounts for 13.7% of everything collected and returns three relevant posts. `cybersecurity` alone accounts for 30.8% of all relevant posts: the corpus time series is, to first order, one search term’s time series, and nothing monitors it. None of the 34 terms names Canada, so any Canadian cyber post not phrased with one of them is unreachable by construction.

## Replacing the relevance gate

The old decision was a conjunction of two narrow lexicons, so recall was bounded by their intersection: 650 of 91,551 posts passed and the 89,714 rejected had never been examined by anything semantic. The replacement poses two *independent* predicates rather than one conjunction — a single combined prompt causes a 4B model to collapse the conjunction onto the side it knows better, admitting global threat-intelligence noise at 6.8% against a true Canadian rate of 1.1%.

| Relevance gate                                    | recall<br>@ registry | specificity<br>@ registry | corpus<br>relevant |
|---------------------------------------------------|----------------------|---------------------------|--------------------|
| <code>rules_v2</code> — keyword conjunction       | 46.5%                | 99.8%                     | 650                |
| <code>llm_v2c</code> — two questions + validation | <b>68.3%</b>         | 99.4%                     | <b>2,348</b>       |

Of the 1,799 posts newly admitted, 84.0% had been rejected on the Canadian side of the conjunction. Two results here are method rather than number. Entity offsets are computed with `str.find` rather than requested from the model, on the narrow grounds that small models miscount them — which turned into a free hallucination detector, since an entity that cannot be located in the post was invented, and nearly half of extracted `place` entities failed it. And the residual error is neither a prompt nor a capability problem: the posts still missed carry no country marker at all (“New post from #Qilin: Coldfish Seafood”), so closing it needs a gazetteer of Canadian organisations, not a larger model.

## The external reference, and the central result

Until this phase every precision and recall figure traced to a single author who wrote the rules, the prompts, and the hand-coded evaluation set. A reported  $\kappa = 0.870$  measured agreement under those conditions, not independence. Ingesting a public ransomware victim registry (1,080 events, 1,943 entities) supplies labels no part of the system produced.

For every Canadian event in the window, the victim organisation’s name was also searched directly on the platform, with no cybersecurity term attached, and every hit passed through the cyber judgement before counting:

| Canadian ransomware events, 22 Aug – 22 Sep 2026  | Events   | Share     |
|---------------------------------------------------|----------|-----------|
| Listed in the registry                            | 27       | —         |
| At least one cyber-related post exists on Bluesky | 26       | 96%       |
| At least one such post was collected by ccint     | 26       | 96%       |
| At least one came from a <b>non-feed</b> account  | <b>2</b> | <b>7%</b> |

The 144 relevant posts come from 15 accounts. Twelve are automated threat-intelligence feeds; the three that are not are Canadian news organisations contributing five posts between them, all about a single event. Individual users contributed nothing; 94.9% of the posts have no likes.

**What looks like coverage is syndication.** The collector is not the bottleneck here — it captured 26 of the 26 events the platform carried. No ransomware trend is detectable because there is almost no ransomware *discussion* to detect: the corpus is a near-complete mirror of several leak-site feeds. That also reframes the power analysis — 8× of this corpus is 8× as much syndication, so volume is the wrong lever.

The one exception is instructive. The Commission de la construction du Québec, listed by the Qilin group with roughly 350,000 records taken, drew genuine Canadian media coverage in both languages. Four of the five posts are in the corpus; the fifth, a radio interview with the CCQ’s chief executive, carries no term from the 34-term list because *protection préventive* is in no lexicon — the collection-boundary gap at human scale.

## A methodological result about evaluation

The prompt revision that fixed recall also generalised “a .ca domain counts” into “a domain counts”, admitting `trustedsec.com`, the UK’s NCSC and `ia.cr` — the Costa Rican TLD — and inflating the corpus from 906 to 3,528 posts, a third of them spurious. The external negative set reported 99.0% specificity throughout, because it is built from formulaic English feed posts about foreign victims: no URLs, no French, and so none of the model’s actual failure modes. **An external label removes self-reference but does not confer representativeness, and a non-representative evaluation set does not merely fail to catch a regression — it certifies one.**

## What the project is, and what comes next

The original framing — trend analysis of Canadian cyber discourse — is not available at this volume for this event class, and adding data will not make it so. What is available, and appears to be unreported in the surveyed literature, is a sensor that states its own blind spots numerically. Published systems report how many events they detected (84 of 151 in the strongest comparable work); none reports how many events had no platform presence at all, because establishing that requires the per-event entity-directed retrieval used here, affordable only when the event count is small. This project’s greatest liability — a corpus of a few thousand relevant posts — is precisely what makes the denominator measurable.

Immediate priorities, in order:

1. **Widen the registry beyond ransomware** — advisories, KEV, Canadian media. Every figure above is conditioned on a source that sees only leak-site victims: the narrowest limitation in the work.
2. **Feed the registry back into the relevance decision.** The residual failure is missing knowledge of Canadian organisation names, and the registry is a list of exactly those names.
3. **Add a Canada-side query stream** as an additional disjunctive stream, not a filter, to reach the population the registry cannot see.
4. **Have an independent annotator label the 200-post sample** already drawn and left blank — the one step that cannot be automated.

*Corpus 91,551 posts, 22 Aug – 22 Sep 2026; registry retrieved 22 Sep 2026; four coexisting annotation versions; 243 tests passing. Methodology in the project wrap-up, coverage measurements in the iteration report.*